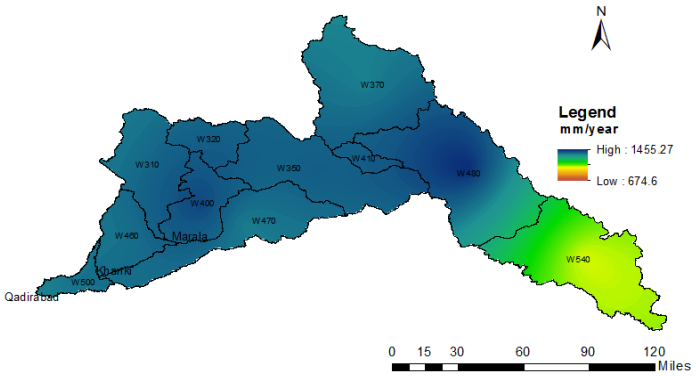


Figure 3.8 IMD4 Average annual rainfall (2003-2014)

3.2.6. River discharges

During flows in July contribution of snowmelt reaches 40% of the total flow (Awan *et al*., 2003). However, peak discharges in the river are observed in August and September. In the monsoon season, two tributaries Jammu Tawi and Munawar Twai, which join Chenab before Marala contributes to most of the discharge in the river. Therefore, these two tributaries play an important role in the overall amount of flood discharge at downstream of Marala (Awan *et al*., 2003).

The discharge data from the following two different locations were considered: Marala head works, where Chenab River enters Pakistan and Qadirabad. The data of Marala was used to calibrate and validate the hydrological model. The second location, Qadirabad barrage, is located almost 80 km downstream of Marala. This data was used for building a hydrodynamic model. Punjab Irrigation department has the responsibility for discharge measurement at these locations and data was gathered from the same agency. Figure 3.9 is showing the location of these gauging stations. Figure 3.10 is showing the daily discharges at Marala from 2003 to 2014**.**

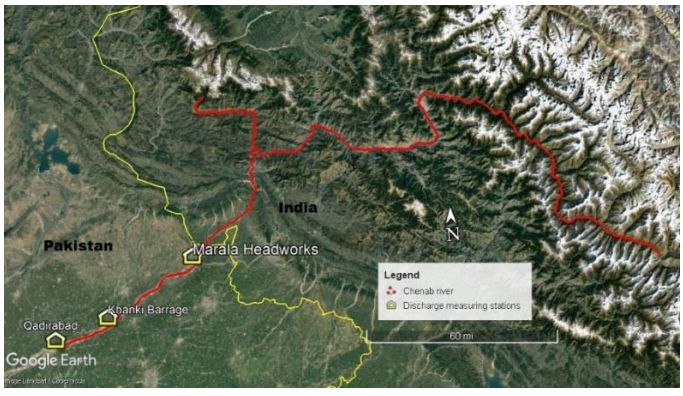


Figure 3.9 Position of discharge gauges in Chenab catchment